Step 4 : Calculation

Calculate the value of the test statistic as

$$\chi_0^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

where, $a = 180$, $b = 60$, $c = 120$, $d = 140$, $N = 500$

$$\chi_0^2 = \frac{500(180\times140-60\times120)^2}{(180+60)(120+140)(180+120)(60+140)} = 43.269$$

Step 5 : Critical value

From chi-square table the critical value at 5% level of significance is $\chi^2_{1,0.05} = 3.841$

Step 6 : Decision

The calculated value $\chi_0^2 = 43.269$ is greater than the critical value $\chi^2_{1,0.05} = 3.841$, the null hypothesis H_0 is rejected. Hence, education of fathers and intelligence of female students are not independent.

2.2.6 Tests for Goodness of Fit

Another important application of chi-square distribution is testing goodness of a pattern or distribution fitted to given data. This application was regarded as one of the most important inventions in mathematical sciences during 20th century. Goodness of fit indicates the closeness of observed frequency with that of the expected frequency. If the curves of these two distributions do not coincide or appear to diverge much, it is noted that the fit is poor. If two curves do not diverge much, the fit is fair.

Computational steps for testing the significance of goodness of fit:

Step 1 : Framing of hypothesis

Null hypothesis H_0 : The goodness of fit is appropriate for the given data set

Alternative hypothesis H_1 : The goodness of fit is not appropriate for the given data set

Step 2 : Data

Calculate the expected frequencies (E_i) using appropriate theoretical distribution such as Binomial or Poisson.

Step 3 : Select the desired level of significance α

Step 4 : Test statistic

The test statistic is

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

where $k =$ number of classes

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